Andrew McKnight

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Education

➤ **Old Dominion University** · 2013 · M.Sc., Dept. of Computer Science

- Thesis-track research in bioinformatics, robotics and 3D image meshing applied to protein folding
- Three published papers with two conference presentations, winning multiple awards
- C++ programming teaching and lab assistant for three semesters

➤ **Old Dominion University** · 2012 · B.Sc., Computer Science

- Graduate-level versions of courses as part of the Accelerated Master's program
- Minor in actuarial mathematics, up to advanced probability, diff-eq and multivariate calculus

Publications

- 2013 · *Protein Loop Length Estimation from Medium Resolution CryoEM Images (M.Sc. thesis)* [Personal Mirror]
- 2013 · *Estimating loop length from CryoEM images at medium resolutions* [Personal Mirror]
- 2013 · *Estimating Lower Bounds on the Length of Protein Polymer Chain Segments using Robot Motion Planning* [Personal Mirror]
- 2012 · *CryoEM Skeleton Length Estimation using a Decimated Curve* [Mirror 1] [Personal Mirror]

Experience

➤ **Sentry** · Jan 2021 – Current · Senior Software Engineer · Remote · FTE

- **Stack**: Objective-C++, Swift, Python, SwiftUI, Django, Claude Code, SPM, XcodeGen, CocoaPods, Carthage, Swiftlint, clang-format, CodeCov, pre-commit, Fastlane, Arkana, GitHub Actions, Redash, SauceLabs, TestFlight, Linear
- **Deliverables**
  - Sentry Cocoa SDK: mobile app crash and performance monitoring
    - Integrated Specto's profiler into Sentry's Cocoa SDK and iterated with new API/features
    - Built User Feedback widget, the first user-facing UI shipped in Sentry Cocoa SDK
    - Built and maintained sample and demo apps for developers and Sales team
    - Parallelized long-running CI jobs and modularized test suites to find and fix statefulness issues
  - Seer for iOS and macOS: experimental apps interacting with our agentic offerings
  - SaaS Billing Platformization: helped overhaul the company's internal billing codebase to help product teams self-serve how they charge for their product segment

➤ **Specto** · Nov 2019 – Jan 2021 · Senior Software Engineer · Remote · FTE

- **Stack**: Objective-C++, Swift, Go, Python, Protobuf, Jinja, XcodeGen, Bazel, GitHub Actions
- **Highlights**
  - First employee at startup; grew and maintained iOS profiling SDK
  - Joined Sentry as acquisition in December 2020; SDK integrated into Sentry Cocoa
- **Deliverables**
  - Specto SDK: mobile app profiling SDK
    - Shared C++ core between iOS and Android
    - Python scripts to generate scoped Xcode projects from Bazel targets (pre `rules-xcode`)
    - Added "safe init" to prevent crashloops, and launch profiling
    - Signal and mach exception propagation with open source contributions to other crash reporters
    - Smoke test generation validating integration across CocoaPods, Carthage, Bazel and SPM

➤ **Rightpoint Consulting** · Jan – Oct 2019 · Senior Software Engineer · Remote · Contract-to-Perm

- **Stack**: Swift, SwiftGen, SwiftLint, Sourcery, CircleCI, Postman, Browserstack, Countly
- **Deliverables**: Identity security app and SDK (under NDA)

➤ **Insight Therapeutics** · Nov 2011 – Jun 2019 · Lead iOS Developer · Norfolk, VA & Remote · Contract

- **Stack**: Objective-C, Swift, MessageUI, Storyboards, CoreGraphics, CoreData, FMDB, SQLite, SSZipArchive, RNCryptor, cocoapods-keys, Trello, BitBucket
- **Highlights**
  - Sole consulatation and development of multiple apps
- **Deliverables**
  - ClotRx: reference e-pamphlet about anticoagulants
    - Inherited from previous developer and extended by me
    - Hybrid app with UIWebView, modernized by me to use WKWebView
    - Added search and bookmark features
    - Added interactive clinical surveys with print/email report generation
  - DementiaRx: reference e-pamphlet about dementia
    - Automatic PDF extraction from original paper publication
    - New app created by me, including search, bookmarking and interactive elements which were then backported to the existing ClotRx app
  - Antibiotic Kinetics: antibiotic dosage calculator
    - Ported from another developer's preexisting Android app
  - eConsent: HIPAA-compliant informed consent app for clinical trial participants
    - Interactive quiz-based presentation of study details
    - Signature collection and export to PDF
    - Interface with backend API to upload finished consents
    - Internal deployment only using enterprise signing
    - Admin mode for managing database of completed documents

➤ **121nexus** · Jan – Apr 2018 · Software Engineer · Remote · Contract

- **Stack**: Swift, AVFoundation, OpenCV, Firebase, cocoapods-keys, AWS
- **Deliverables**
  - 121 Scan: Medical inventory management app/SDK with custom barcode/QR scanning
    - Public launches via iTunes and a white-labeled internal enterprise app
    - Reimplemented major area of app with new UI and platform API
    - Added crash reporting (Crashlytics) and logging (XCGLogging)
    - Augmented and documented analytics (Firebase)

➤ **Layer** · Dec 2016 – Jul 2017 · Software Engineer · Remote · FTE

- **Stack**: C, Objective-C, Swift, OCMock, Specta, Expecta, Rails, Thrift, WebSockets, SPDY, SQLite, CocoaPods, Jenkins
- **Highlights**
  - Rearchitected test suite to separate unit tests from integration/end-to-end tests
  - Streamlined developer environment configuration (Rake, Gems, CocoaPods and git submodules)
  - Migrated old SPDY/Thrift architecture to WebSockets with rewritten Thrift specs
  - Rearchitected auth and transport layers into state machines using TransitionKit
- **Deliverables**: SDKs for data and UI components, an embeddable app layer and fully-functional sample app

➤ **Twitter** · Apr 2015 – Dec 2016 · Software Engineer · Cambridge, MA · FTE

- **Stack**: Objective-C, C, Swift, Java, Scala, Clojure, Coffeescript, Rails, React, Protobuf, Storm, Heron, Cascading, Scalding, Graphite, Beanstalk, Kafka, Redis, Cassandra, AWS, TeamCity, Chef, Trello, Vertica, Zeppelin
- **Highlights**
  - Presented work on rearchitectures at MobileDevCon 2016 in Amsterdam
  - Maintenance of backend lambda architecture, including on-call rotations
- **Deliverables**
  - Crashlytics SDK: Crash reporting SDK for Apple platforms
    - Rearchitected CLI tools that ship inside the framework for bootstrapping projects and uploading dSYMs
    - Added Bitcode and tvOS compatibility to SDKs, and investigated watchOS support
    - Pioneered new backend feature to cross-reference common crashes across apps and present developers with suggested fixes
  - Fabric SDK and macOS app
    - Assisted adding Unity iOS support
    - Streamlined build and release processes with TeamCity agents deployed via Chef
    - Rearchitected startup procedure using NSOperation, including custom async and compound operation implementations
  - FABConfig: Standardizes config of all Xcode projects and released the open source FABConfig
  - FABOperation: Custom NSOperation subclasses for async work and grouping tasks

➤ **MC10** · Jan – Apr 2015 · Software Engineer · Cambridge, MA · FTE

- **Stack**: Objective-C, Swift, CoreBluetooth, NSOperation, TeamCity, Fabric
- **Highlights**
  - Moved all existing demo code into Git repos, no previous SCM
  - Architected SDK and app codebases from the ground up
  - Set up continuous integration and deployment with TeamCity and Fabric
  - Assisted adoption of Agile processes, including delivering internal presentations
- **Deliverables**: Companion iOS app for clinical research wearable device

➤ **Raizlabs** · Jul 2013 – Dec 2014 · Software Engineer · Boston, MA · FTE

- **Stack**: Objective-C, Swift, Jenkins, TeamCity
- **Deliverables**
  - Care.com: service seeker/provider marketplace
    - iOS 7 redesign, with perspective zoom effect for a horizontally paged scroll view
    - Dynamic UI generation from server-supplied data
  - B&H Photo: e-commerce app
    - Custom parallactic scrolling of table view headers
    - Push notifications with deep linking
    - UICollectionViews for device-specific layouts on iPhone and iPad
  - UniFirst: interactive enterprise sales catalog on iPad
    - Drag and drop editing of clothing selections on human models
    - Patch designer with logo image, font and layout customization
    - PDF generation for sales orders and reports
  - Invaluable: realtime high-end auction platform on iPad
    - Paging of API result display in table/collection views with custom pull-to-request footer and animation

➤ **ODU Systems Lab** · Oct 2012 – May 2013 · Software Engineer · Norfolk, VA · Student Intern

- **Highlights**
  - Supervised two foreign exchange students
  - Consulting with clients on site
- **Deliverables**
  - ChirpXM: location based social networking app (defunct)
  - Prototyped EMR iPad app for CHKD

➤ **Intergraph** · May – Oct 2012 · Software Engineer · Hampton, VA · Intern

- **Stack**: C#, VisualBasic, SQL, QF-Test, Coverity, Visual Studio, TeamFoundation
- **Highlights**
  - Unit and UI testing, regression testing in numerical modules
  - Manual acceptance testing for new features and fixes

➤ **Old Dominion University** · Aug 2012 – May 2013 · Project co-lead · Norfolk, VA · Capstone Project

- **Stack**: Objective-C, OpenEars, MapQuest
- **Highlights**
  - Co-lead of 7 member team that received highest score in the program
  - 4 presentations given to visiting professors and industry executives
  - Sole developer of iOS component of total solution

➤ **Old Dominion University** · Aug 2012 – May 2014 · Research Assistant · Norfolk, VA · Graduate Student

- **Stack**: Java, C++, CGAL, Boost, Jama
- **Highlights**
  - Image processing, robotics and machine learning applications in bioinformatics
  - Multiple publications and conference presentations

➤ **Old Dominion University** · Aug 2012 – May 2014 · Lab Instructor · Norfolk, VA · Teaching Assistant

- Online delivery of CS 333: Problem solving and programming in C++
- Lab instructor for CS 250: Intermediate programming in C++
- Assistant instructor for CS 150: Introduction to programming in C++
- Website maintenance, grading, office hours

➤ **XVD Longboards** · Oct 2010 – Apr 2011 · Web Developer · Norfolk, VA · Contract

- **Stack**: ActionScript, Flash Studio
- **Highlights**
  - Paid with a couple custom longboards, which I got to help build myself!
- **Deliverables**: Flash-based website to design custom longboards for purchase

Honors & Awards

- 2013 · Gene Newman Award · Virginia Modeling, Analysis and Simulation Center Capstone conference · Best overall paper
- 2012 · Graduate with Distinction · Old Dominion University · Awarded with B.Sc.
- 2012 · Undergraduate Research Award · ODU Spring Undergraduate Research Fair · Outstanding undergraduate achievement
- 2011 · First Place · ODU-ACM Programming Contest
- 2010 – 2011 · Dean's List · Old Dominion University
- 2009 · Summa Cum Laude · Piedmont Virginia Community College

Presentations

➤ **iOSDevCampCO** · May 2018 · *Colloquial Swift* (45 min) · Colorado Springs, CO

- Presented findings from analysis of common function implementations in open source repositories as a way to generate possible Swift Evolution proposals

➤ **CocoaHeads Boston** · Jun 2017 · *UIViewController Decomposition and UI Reuse* (20 min) · MIT, Cambridge, MA

- Discussing strategies to separate responsibilities from UIViewControllers and a library of reusable UI components, e.g. animated transparent modal presentation with a close button

➤ **Boston Algorithms Meetup** · Jan 2017 · *HyperLogLog* (20 min) · Twitter, Cambridge, MA

- Helped organized this initial meeting of the group
- Covered the theoretical workings of the HyperLogLog algorithm, with a copresenter giving a working demostration

➤ **mDevCon** · Mar 2016 · *Command and NSOperation* (45 min) · Amsterdam, NL

- Delivered presentation on the Command software design pattern, and NSOperation
- Introduced subclasses we open sourced at Crashlytics for asynchronous and compound operations
- Previously delivered at a lunchtime tech talk at Twitter, Cambridge, MA (Nov 2015) and CocoaHeads Boston meetup, MIT, Cambridge, MA (Dec 2015)

➤ **VMASC Capstone Conference** · 2013 · *Estimating lower bounds on the length of protein polymer chain segments using robot motion planning* · Suffolk, VA

- Presented research on using computational geometry and motion planning algorithms to model protein molecule structures
- Best paper award

➤ **IEEE WCSB** · Oct 2012 · *CryoEM skeleton length estimation using a decimated curve* · Philadelphia, PA

- Presented research on using a dynamic graph algorithm to identify protein molecule substructures from 3D images

Mentorship

➤ **Girls Who Code** · Jul 2016 · *iOS Dev Session* · Twitter, Cambridge, MA

- Part of GWC summer internship covering programming first principles to full-stack style development
- Designed and led iOS session where we built a Flappy Bird clone, introducing them to Xcode as an IDE, Interface Builder, Playgrounds and Swift

➤ **NeighborNest** · Oct 2015 · *Web Dev Session* · Twitter, San Francisco, CA

- Community program to help advocate technology literacy
- Assisted in session where students worked on a Twitter clone, teaching functional programming concepts, Javascript, and some front- and back- end concepts